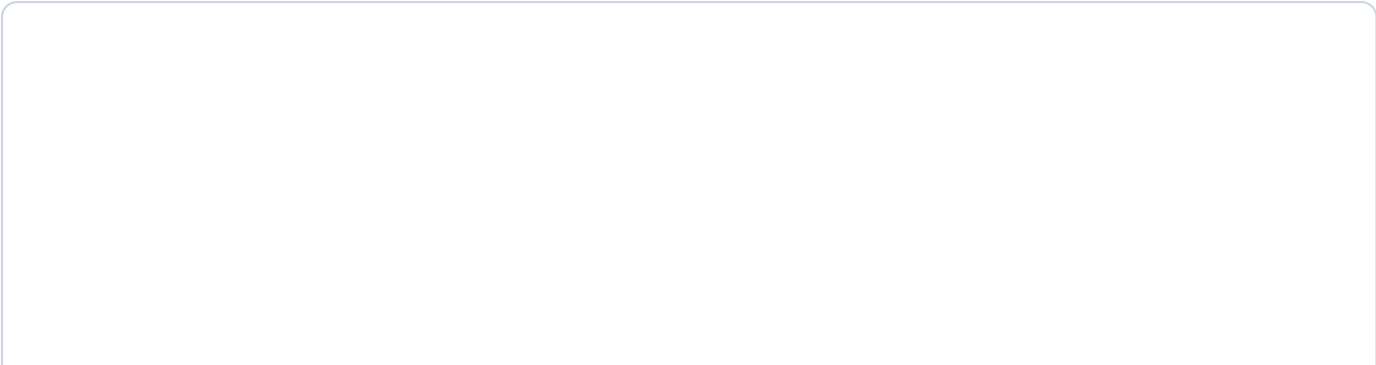
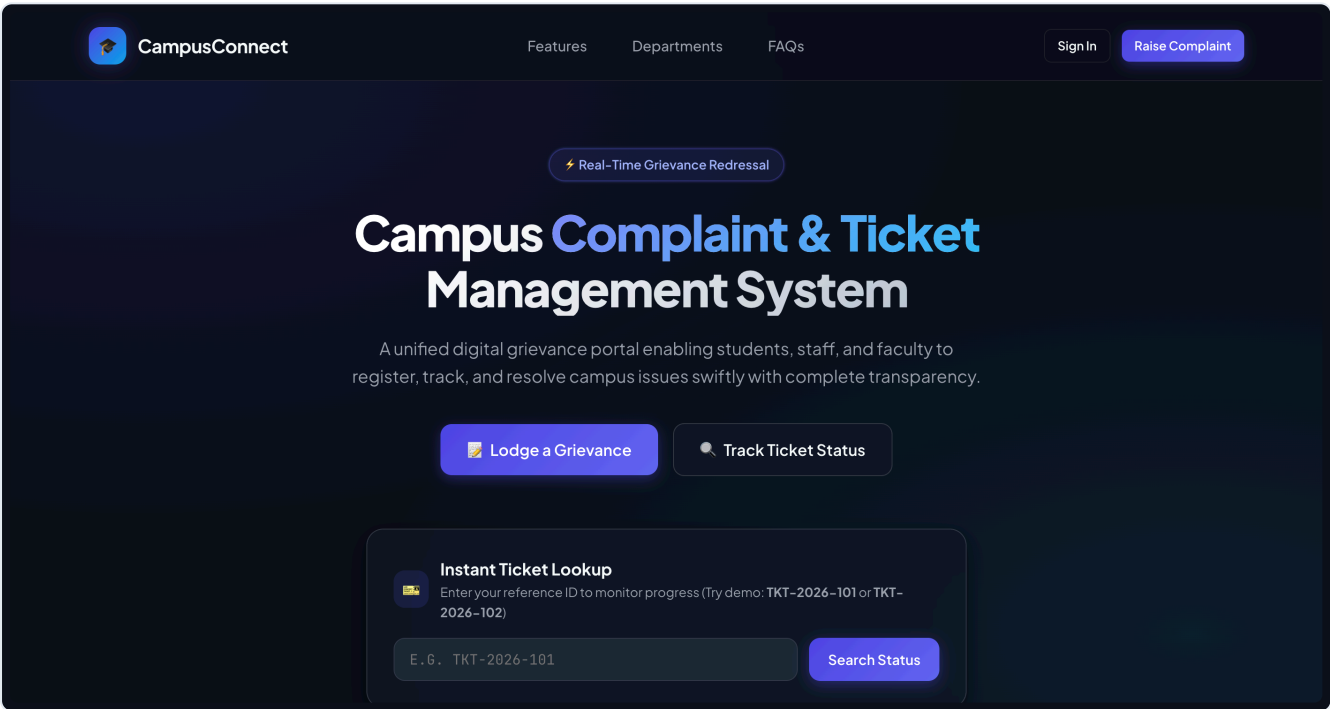


PROJECT: CampusConnect Portal	PHASE: Part 1 (Frontend & DB Design)	STACK: HTML5, Vanilla CSS3, ES6 JS	DATABASE: MySQL 3NF (InnoDB)
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Output 1: Public Landing Page & Instant Status Lookup

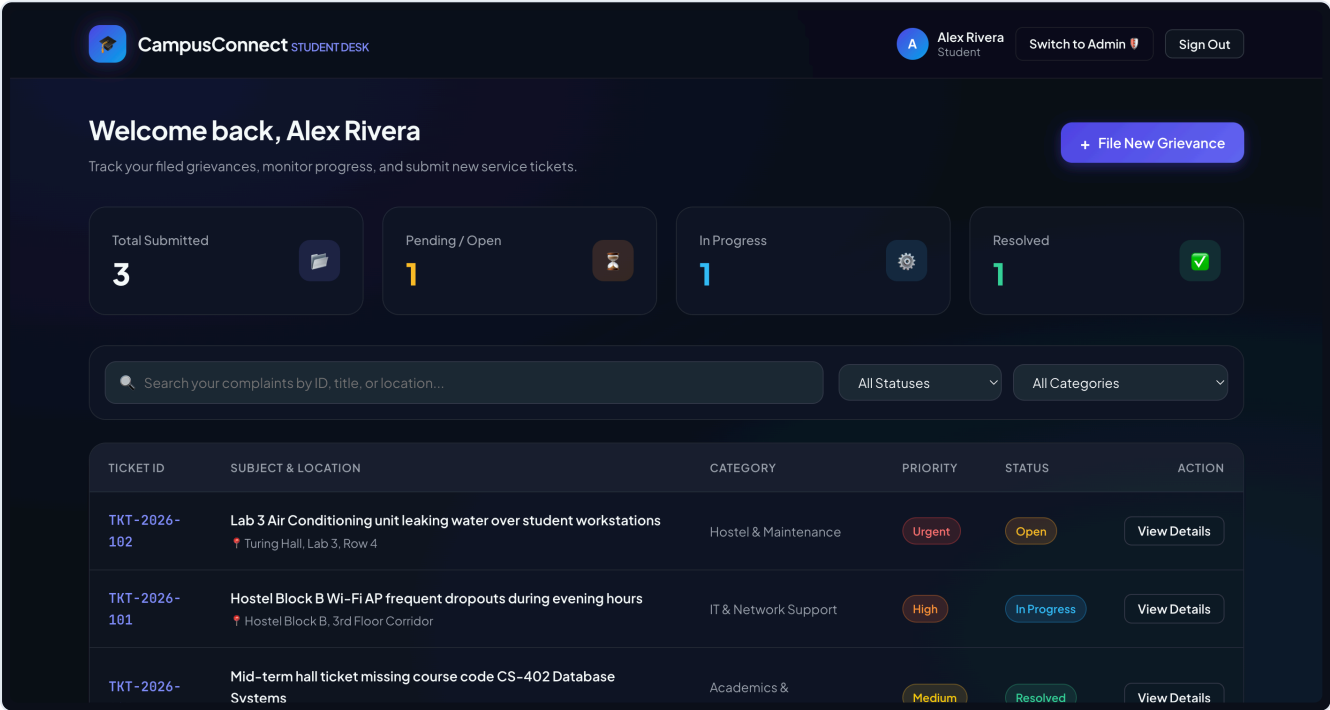
FRONTEND • INDEX.HTML



Description: The public-facing landing page features a modern dark glassmorphism design system, branded navigation, feature highlights, and an **Instant Ticket Lookup** widget. Students and staff can track any filed grievance in real time by entering their reference ID (e.g. TKT-2026-101) without prior authentication.

Output 2: Student Grievance Management Portal

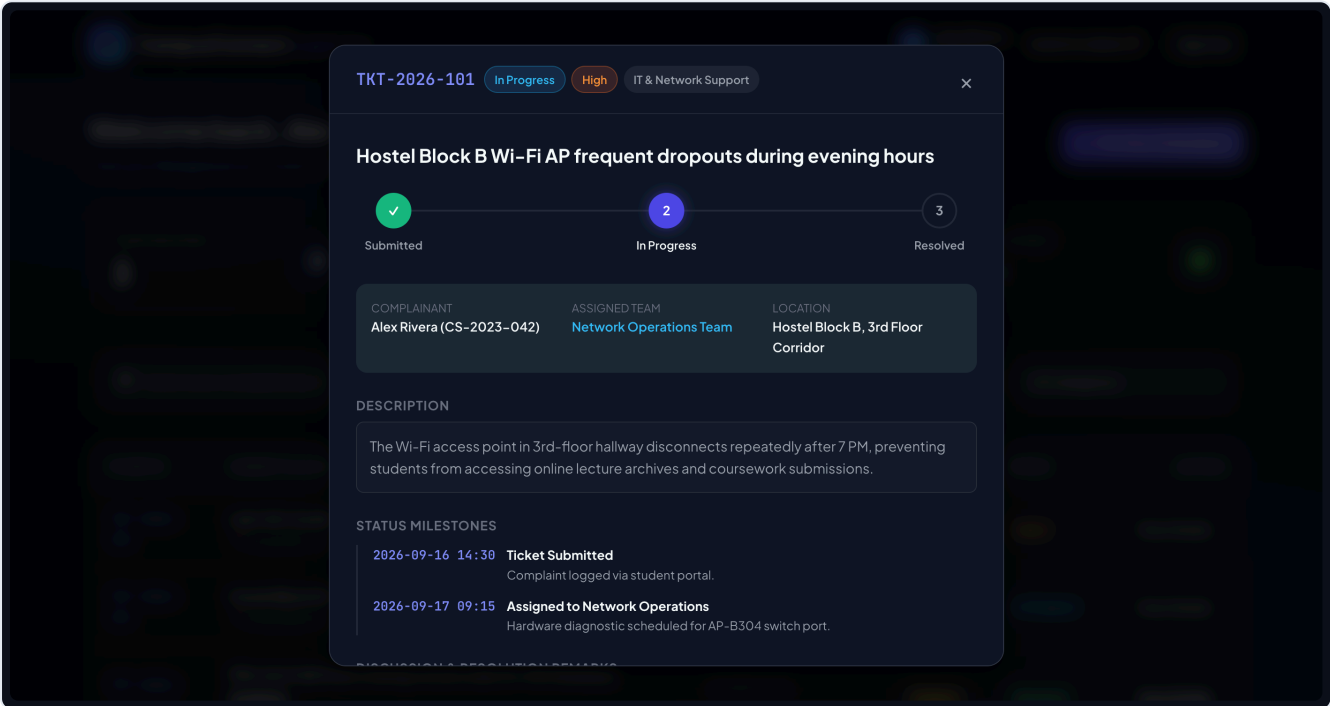
FRONTEND • USER-DASHBOARD.HTML



Description: The authenticated student portal displaying active grievances for student **Alex Rivera (CS-2023-042)**. It provides real-time KPI metrics (*Total Submitted, Pending/Open, In Progress, Resolved*), dynamic keyword search, departmental category filters, status badges, and a direct modal trigger to lodge new campus complaints.

Output 3: Grievance Lifecycle Stepper, Audit Trail & Discussion

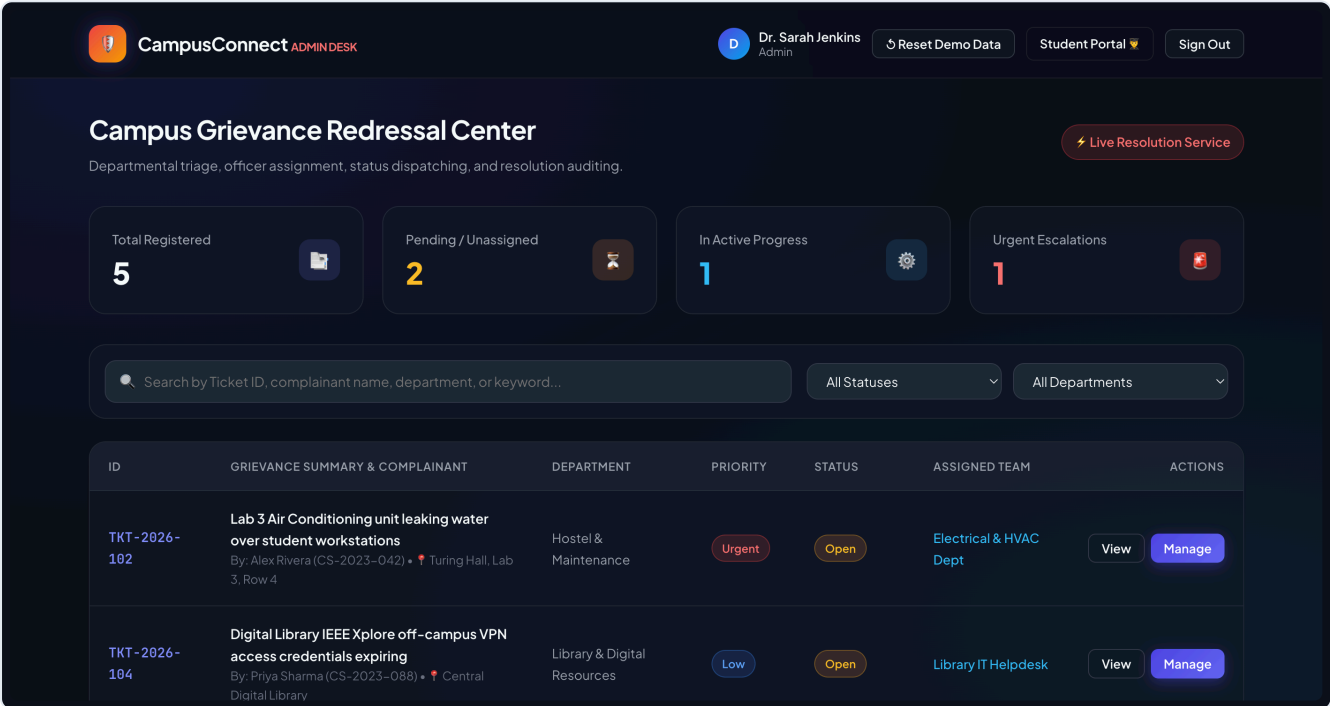
FRONTEND • INTERACTIVE MODAL



Description: Interactive ticket inspection dialog for TKT-2026-101 showcasing end-to-end resolution transparency. Includes a **visual 3-stage progress stepper** (*Submitted* → *In Progress* → *Resolved*), assigned response department (*Network Operations Team*), chronological audit trail milestones, and a live bidirectional discussion thread.

Output 4: Administrative Grievance Redressal & Triage Center

FRONTEND • ADMIN-DASHBOARD.HTML



Description: Administrative triage console operated by Chief Grievance Officer **Dr. Sarah Jenkins**. Displays real-time institutional grievance counters, urgency highlights, and master queue controls. Administrators can filter by department, review complainant student details, dispatch technicians, and manage resolution statuses.

Output 5: Relational Database Schema Design (3NF RDBMS)

DATABASE DESIGN • CAMPUS_CONNECT_SCHEMA.SQL

CampusConnect — Database Schema

campus_connect_schema.sql
MySQL / MariaDB - Relational schema for the complaint & ticket management system

```
1 -- =====  
2 -- CampusConnect: Complaint & Ticket Management System  
3 -- Relational Database Schema (MySQL / MariaDB system)  
4 -- Designed from the existing frontend data model in js/store.js  
5 -- =====  
6  
7 CREATE DATABASE IF NOT EXISTS campus_connect;  
8 USE campus_connect;  
9  
10 -- =====  
11 -- 1. USERS -- base table for every account (student or admin)  
12  
13 CREATE TABLE users (  
14   user_id INT AUTO INCREMENT PRIMARY KEY,  
15   name VARCHAR(100) NOT NULL,  
16   email VARCHAR(100) NOT NULL UNIQUE,  
17   password_hash VARCHAR(255) NOT NULL,  
18   role ENUM('student', 'admin') NOT NULL,  
19   phone VARCHAR(20),  
20   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
21 );  
22  
23 -- =====  
24 -- 2. STUDENTS -- 1:1 extension of users where role = 'student'  
25  
26 CREATE TABLE students (  
27   user_id INT PRIMARY KEY,  
28   student_id VARCHAR(100) NOT NULL UNIQUE, -- e.g. CS-2023-042  
29   department VARCHAR(100) NOT NULL,  
30   FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE  
31 );  
32  
33 -- =====  
34 -- 3. ADMINS -- 1:1 extension of users where role = 'admin'  
35  
36 CREATE TABLE admins (  
37   user_id INT PRIMARY KEY,  
38   department VARCHAR(100) NOT NULL, -- e.g. Campus Grievance & Operations Cell  
39   title VARCHAR(100),  
40   FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE  
41 );  
42  
43 -- =====  
44 -- 4. CATEGORIES -- lookup table (IT & Network Support, Hostel & Maintenance, ...)  
45  
46 CREATE TABLE categories (  
47   category_id INT AUTO INCREMENT PRIMARY KEY,  
48   name VARCHAR(100) NOT NULL UNIQUE  
49 );  
50  
51 -- =====  
52 -- 5. TICKETS -- the core grievance/ticket record  
53  
54 CREATE TABLE tickets (  
55   ticket_id VARCHAR(200) PRIMARY KEY, -- e.g. TKY-2026-101  
56   title VARCHAR(200) NOT NULL,  
57   description TEXT NOT NULL,  
58   category_id INT NOT NULL,  
59   priority ENUM('low', 'medium', 'high', 'urgent') NOT NULL DEFAULT 'medium',  
60   status ENUM('open', 'in progress', 'resolved', 'closed') NOT NULL DEFAULT 'open',  
61   location VARCHAR(200),  
62   submitted_by INT NOT NULL, -- PK to users (student who filed it)  
63   assigned_to VARCHAR(100) DEFAULT 'Pending Assignment', -- staff/department name  
64   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
65   updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,  
66   FOREIGN KEY (category_id) REFERENCES categories(category_id),  
67   FOREIGN KEY (submitted_by) REFERENCES users(user_id)  
68 );  
69  
70 -- =====  
71 -- 6. TICKET TIMELINE -- status history / audit trail per ticket  
72  
73 CREATE TABLE ticket_timeline (  
74   timeline_id INT AUTO INCREMENT PRIMARY KEY,  
75   ticket_id VARCHAR(200) NOT NULL,  
76   status ENUM('open', 'in progress', 'resolved', 'closed') NOT NULL,  
77   label VARCHAR(100) NOT NULL, -- e.g. 'Assigned to Network Operations'  
78   event_date DATETIME NOT NULL,  
79   note TEXT,  
80   FOREIGN KEY (ticket_id) REFERENCES tickets(ticket_id) ON DELETE CASCADE  
81 );  
82  
83 -- =====  
84 -- 7. TICKET COMMENTS -- conversation thread on a ticket  
85  
86 CREATE TABLE ticket_comments (  
87   comment_id INT AUTO INCREMENT PRIMARY KEY,  
88   ticket_id VARCHAR(200) NOT NULL,  
89   author_id INT NOT NULL, -- FK to users (student or admin who commented)  
90   role ENUM('student', 'admin') NOT NULL,  
91   comment_text TEXT NOT NULL,  
92   comment_date DATETIME NOT NULL,  
93   FOREIGN KEY (ticket_id) REFERENCES tickets(ticket_id) ON DELETE CASCADE,  
94   FOREIGN KEY (author_id) REFERENCES users(user_id)  
95 );  
96  
97 -- =====  
98 -- Seed data: categories (matches the dropdowns in user-dashboard.html)  
99  
100 INSERT INTO categories (name) VALUES  
101 ('IT & Network Support'),  
102 ('Hostel & Maintenance'),  
103 ('Academic & Examination'),  
104 ('Library & Digital Resources'),  
105 ('Catering & Mess'),  
106 ('Accounts & Finance');  
107  
108 -- =====  
109 -- Useful indexes for the queries the frontend already runs  
110 -- (filter by student email, status, category, priority, search)  
111  
112 CREATE INDEX idx_tickets_status ON tickets(status);  
113 CREATE INDEX idx_tickets_priority ON tickets(priority);  
114 CREATE INDEX idx_tickets_category ON tickets(category_id);  
115 CREATE INDEX idx_tickets_submitted_by ON tickets(submitted_by);
```

Generated from campus_connect_schema.sql

Description: Third Normal Form (3NF) relational database schema specification in MySQL / MariaDB (campus_connect_schema.sql & campus_connect_schema.pdf). Features **7 normalized entities** (users, students, admins, categories, tickets, ticket_timeline, ticket_comments) with primary/foreign key constraints, cascading referential actions, and query performance indexes.